Question 1

Consider the SIRI epidemic model, where individuals can be in one of the following states: Susceptible (S), Infected (I), and Recovered (R). The total population size is fixed and given by $N = S + I + R$. The reaction scheme of the model is $S+I \to I+I$, $I \to R$, $R \to I$, with the corresponding transition rates $\frac{\alpha SI}{N}$, βI, γR, where α is the infection rate, β is the recovery rate, and γ is the relapse (loss of immunity) rate.

1. Using the above reaction scheme, write down the Master equation governing the stochastic dynamics of the SIRI model. Clearly explain the physical meaning of each term.
2. Write a pseudocode to numerically simulate the stochastic SIRI dynamics using the Gillespie algorithm.
3. Using the reaction scheme of the SIRI epidemic model, write down the system of ordinary differential equations (ODEs) governing the time evolution of the number of susceptible $S(t)$, infected $I(t)$, and recovered $R(t)$ individuals. Briefly explain the physical meaning of each term appearing in the equations.

   Note: The first reaction scheme($S+I \to I+I$) corresponds to the conversion of a susceptible individual into an infected individual upon contact with an infected one.

(3 + 3 + 3 = 9 marks)

Susceptible — when interacted with infection they will get infected

Infected — The ones who got infected by the disease

Recovred — The ones who are infected & recoved fol

$S+I \to I+I$ rate $\frac{\alpha SI}{N}$

$I \to R$ rate βI $N = S+I+R$

$R \to I$ rate γR

I $(S+1, I-1, R) \to (S, I, R)$

gaining losing

$$= \alpha \frac{(S+1)(I-1)}{N} P(S+1, I-1, R) - \frac{\alpha SI}{N} P(S,I,R)$$

II $(S, I+1, R-1) \to (S, I, R)$

$$= \beta(I+1) P(S, I+1, R-1) - \beta I P(S, I, R)$$

III $(S, I-1, R+1) \to (S, I, R)$ 2

$$= \gamma(R+1) P(S, I-1, R+1) - \gamma R P(S,I,R)$$

we get SIR from that state, losing of initial SIR then no reaction thenalso s:
So, substract

Question 2

A random walker moves in one dimension with step size d_0. At each time step, the walker moves either to the right or to the left with equal probability 1/2. The walker starts from the origin.

1. Show that the probability of taking n_1 steps in one direction (say, to the right) out of a total of N steps follows a binomial distribution.
2. Write the expected value of n_1 (no proof required).
3. Calculate the average displacement of the walker after N steps.
4. Write a pseudocode or algorithm to compute the average displacement.

7.5

(2 + 1 + 2 + 3 = 8 marks)

dom walk

$P = q = 1/2$

He take n_1 steps to the right so $N - n_1$ step to left

$\underbrace{r\ r\ r\ r\ r}_{n_1}\ \underbrace{L\ L\ L\ L}_{N-n_1}$

it has a probability of $p^{n_1} q^{N-n_1}$

d we can distribute it in ~~any~~ way in $\frac{N!}{n_1!(N-n_1)!}$

$$\text{Prob}(n_1 \text{ step in right}) = \frac{N!}{n_1!(N-n_1)!} p^{n_1} q^{N-n_1}$$

$$= \frac{N!}{n_1!(N-n_1)!} \left(\frac{1}{2}\right)^N$$

$\langle n_1 \rangle = NP = \frac{N}{2}$ +1

m be the ~~...~~ displacemet

$m = (n_1 - (N - n_1)) d_0$

$m = (2n_1 - N) d_0$

Question 3

Consider the two-dimensional Lotka–Volterra predator–prey model

$$\frac{dx}{dt} = \alpha x - \beta xy, \qquad \frac{dy}{dt} = \delta xy - \gamma y,$$

where $x(t)$ and $y(t)$ denote the prey and predator populations, respectively, and $\alpha, \beta, \gamma, \delta > 0$ are constants.

1. Identify all equilibrium states of the system.
2. Using eigenvalue analysis, classify the nature and stability of each equilibrium.
3. Comment on the long-term population behavior implied by your results.

(0.5+1+0.5=2 marks)

for equilibrium $\frac{dx}{dt} = 0$ $\quad \frac{dy}{dt} = 0$

$\alpha x - \beta xy = 0 \qquad \delta xy - \gamma y = 0$

$x\left(y - \frac{\alpha}{\beta}\right) = 0 \qquad y\left(x - \frac{\gamma}{\delta}\right) = 0$

$x = 0, \; y = 0 \qquad \cancel{y} \; x = \frac{\gamma}{\delta}, \; y = \frac{\alpha}{\beta}$

$(0,0)$, $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ are the two equilibrium States of system

0.5

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{bmatrix} \alpha - \beta y & -\beta x \\ \delta y & \delta x - \gamma \end{bmatrix}$$

for $(0,0)$

$$J = \begin{bmatrix} \alpha & 0 \\ 0 & -\gamma \end{bmatrix}$$

eigen value $= \alpha, -\gamma$ ✓ 0.5

(unstable) (Saddle)

for $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$

$$J = \begin{bmatrix} 0 & -\frac{\beta\gamma}{\delta} \\ \frac{\delta\alpha}{\beta} & 0 \end{bmatrix}$$

8

$$\left|\begin{matrix} -\lambda & -\frac{\beta\gamma}{\delta} \\ \frac{\delta\alpha}{\beta} & -\lambda \end{matrix}\right| = 0$$

$\lambda^2 + \alpha\gamma = 0$

0.25 $\quad \lambda = \pm i\sqrt{\alpha\gamma}$ ~~(Saddle)~~ ~~(unstable)~~ (Centre)

wrote error

after a long time it behaves like oscillatory

prey ↑ predator ↓ predator ↑ prey ↓

~~One feed on prey~~

~~Prey feed on predator, predator ↓ prey ↑~~

~~predator less so prey decrease~~

~~Prey decreases~~

Predator feeds on prey, Prey ↓ Predator ↑

prey less so predator also decrese

predetor less prey increase, this reapeats again & again

0.25

Question 4

How would you determine whether a given sequence of data is randomly drawn from a uniform distribution and does not exhibit periodicity?

(2 marks)

by SLLN

$$E[x] = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} f(n)$$ (strong law of large numbers)

we have given sequence of data which is randomly drawn from uniform distribution.

Question 5

Consider the function

$$f(x) = \begin{cases} \sin(x)\cos(x), & 0 \leq x \leq \frac{\pi}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

1. Determine the normalization constant and hence write the properly normalized probability density function.

2. Explain how a machine that generates uniform random numbers in the interval $(0, 1)$ can be used to generate random numbers that follow this distribution.

(1+3=4 marks)

$$k\int_0^{\infty} f(x)\,dx = 1$$

$$k\int_0^{\pi/2} SxCx + 0 = 1$$

$$\int_0^{\pi/2} \frac{k}{2}\, S2x\, dx = 1$$

$$\frac{k}{2} \left. \frac{-C2x}{2} \right|_0^{\pi/2} = 1$$

$$\frac{k}{2} - \frac{(-1-(1))}{2} = 1$$

$$\frac{k}{4}(2) = 1 \Rightarrow \frac{k}{2} = 1 \quad k = 2$$